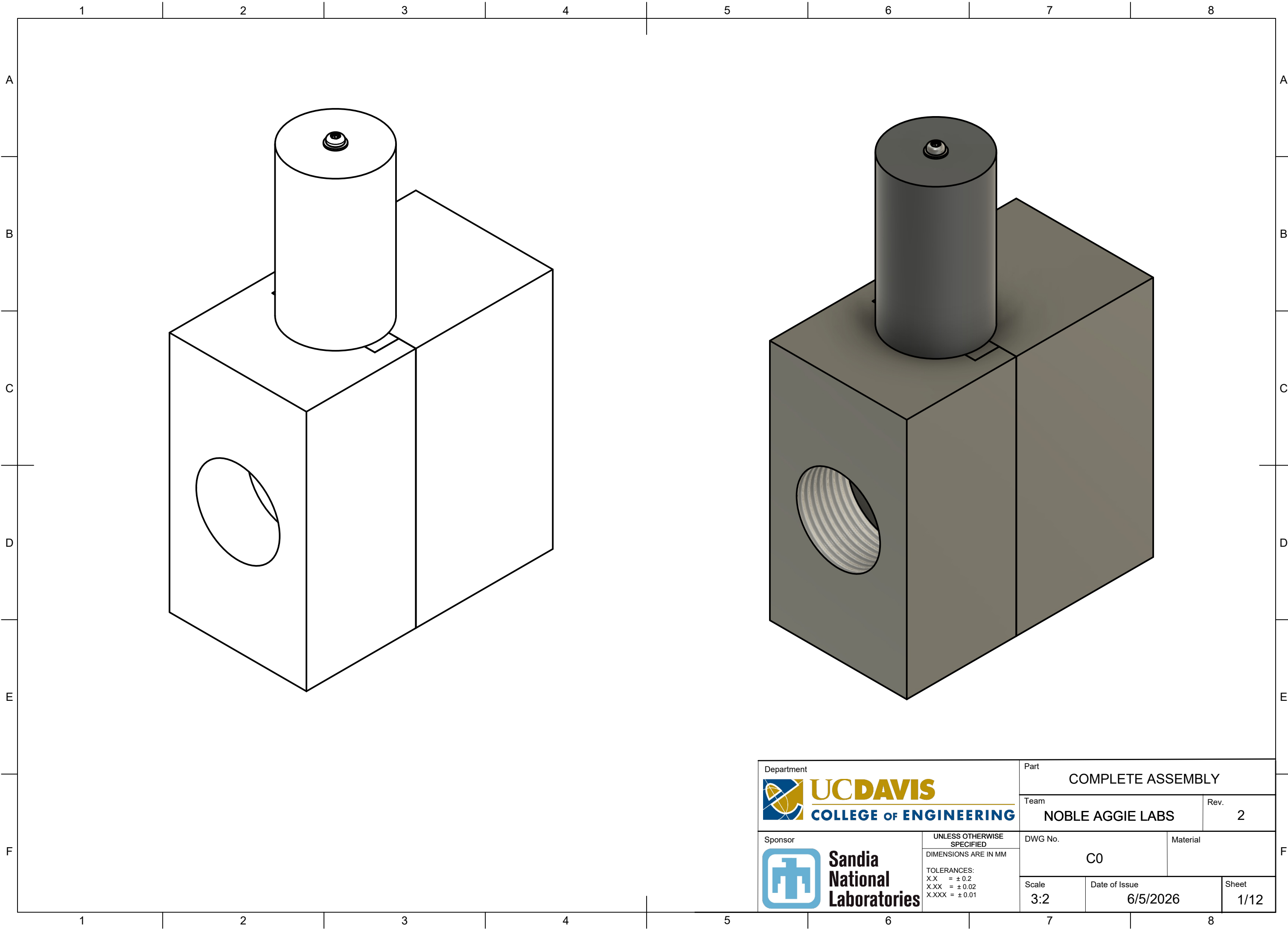
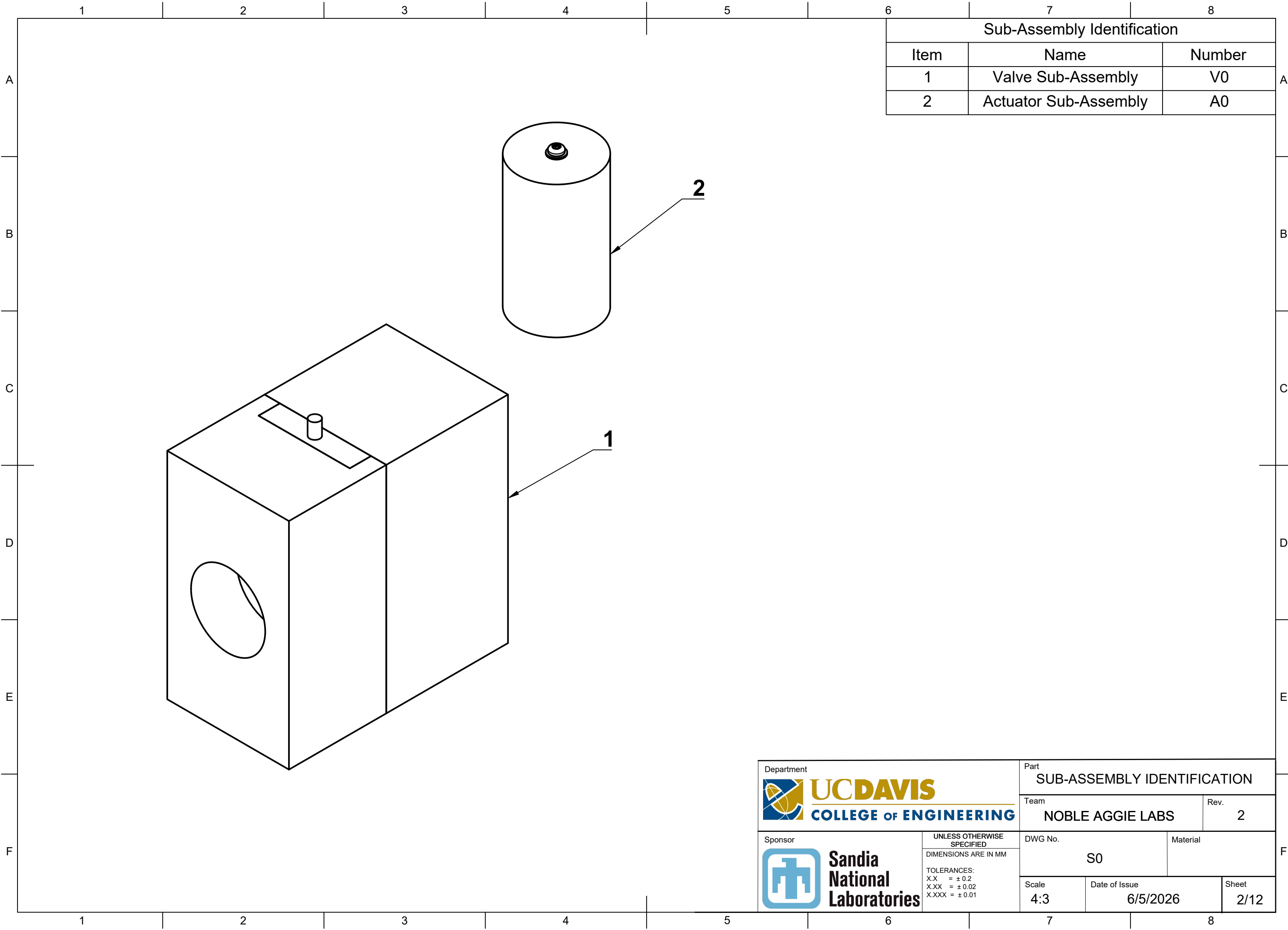


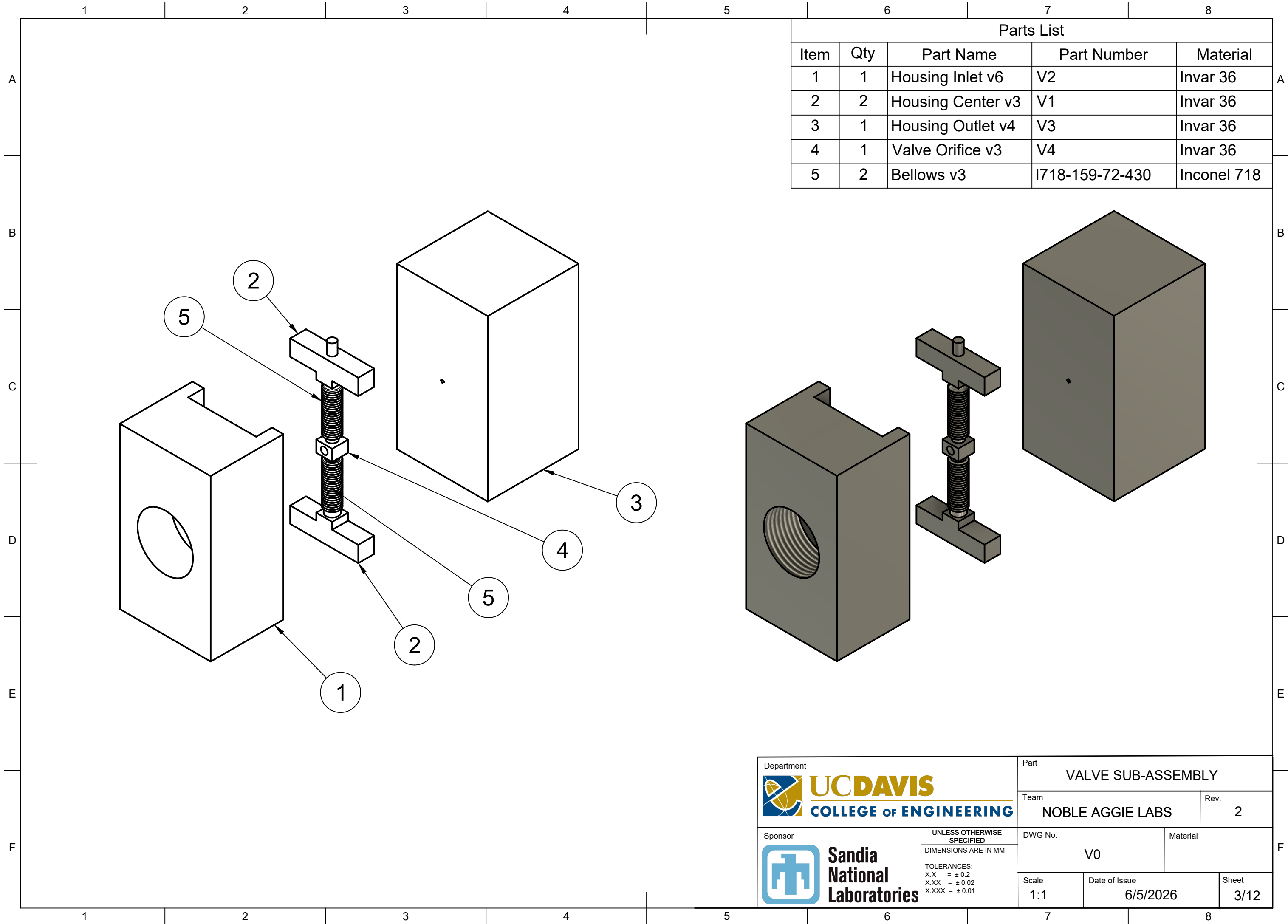


Department		Part		
		COMPLETE ASSEMBLY		
		Team		Rev.
		NOBLE AGGIE LABS		2
Sponsor		DWG No.		Material
		C0		
		Scale	Date of Issue	Sheet
		3:2	6/5/2026	1/12

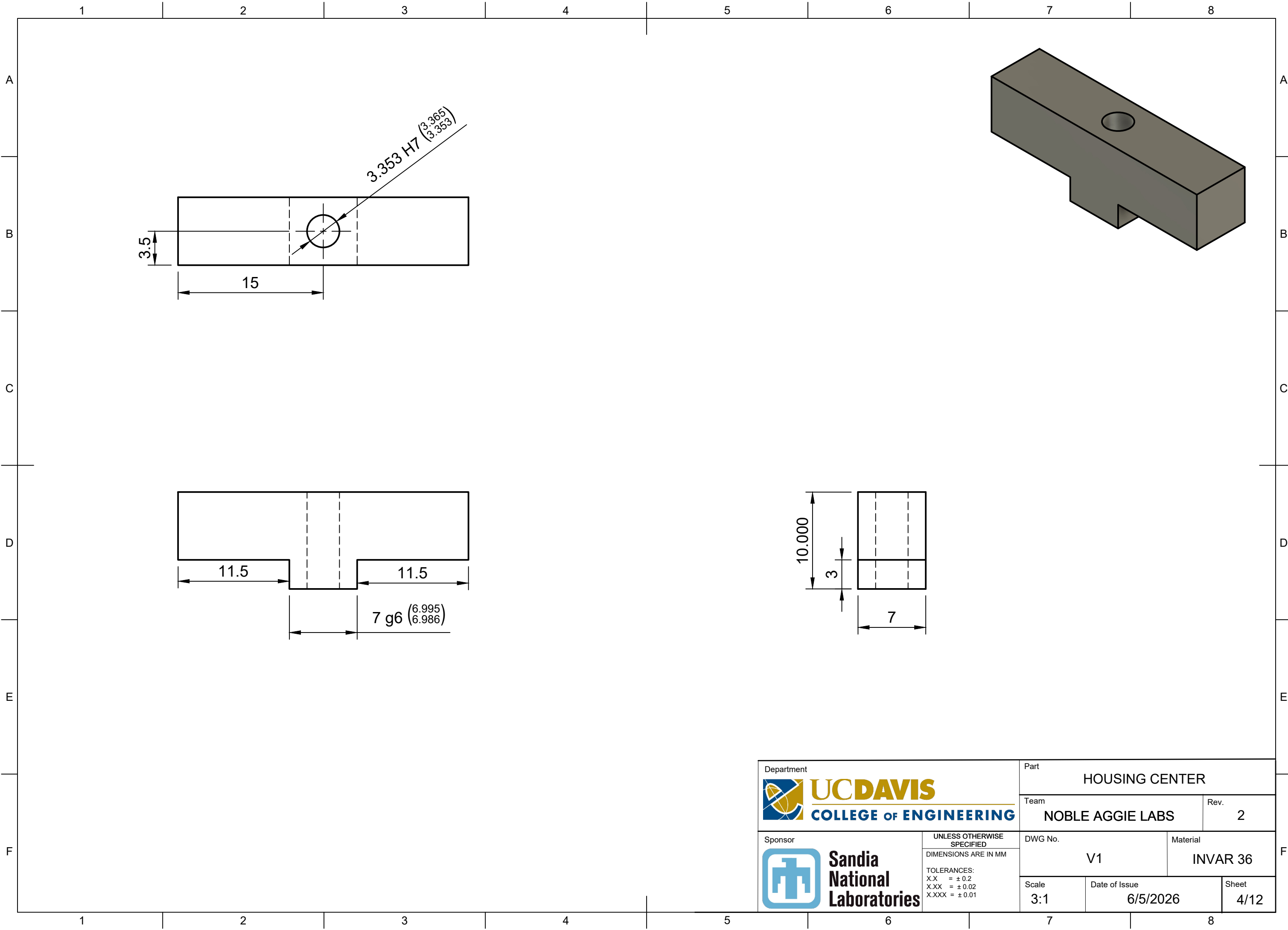


Sub-Assembly Identification		
Item	Name	Number
1	Valve Sub-Assembly	V0
2	Actuator Sub-Assembly	A0

Department		Part	
		SUB-ASSEMBLY IDENTIFICATION	
Team		Rev.	
NOBLE AGGIE LABS		2	
Sponsor		DWG No.	
		S0	
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		Material	
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		Scale	Date of Issue
		4:3	6/5/2026
		Sheet	
		2/12	



Department		Part		
		VALVE SUB-ASSEMBLY		
Sponsor		Team	Rev.	
		NOBLE AGGIE LABS	2	
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.	Material	
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		V0		
Scale		Date of Issue	Sheet	
1:1		6/5/2026	3/12	



Department		Part		
		HOUSING CENTER		
Sponsor		Team	Rev.	
		NOBLE AGGIE LABS	2	
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.	Material	
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		V1	INVAR 36	
		Scale	Date of Issue	Sheet
		3:1	6/5/2026	4/12

1 2 3 4 5 6 7 8

A

B

C

D

E

F

A

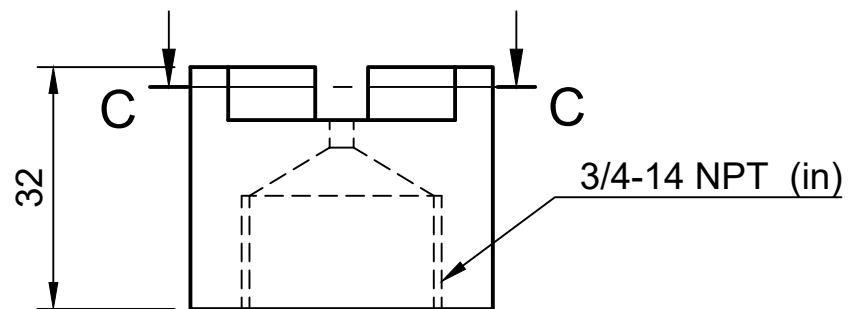
B

C

D

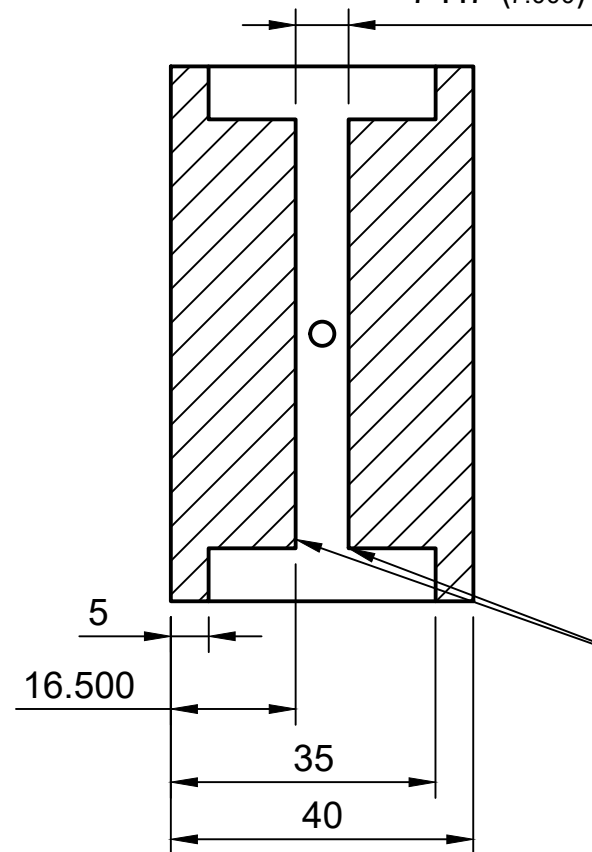
E

F

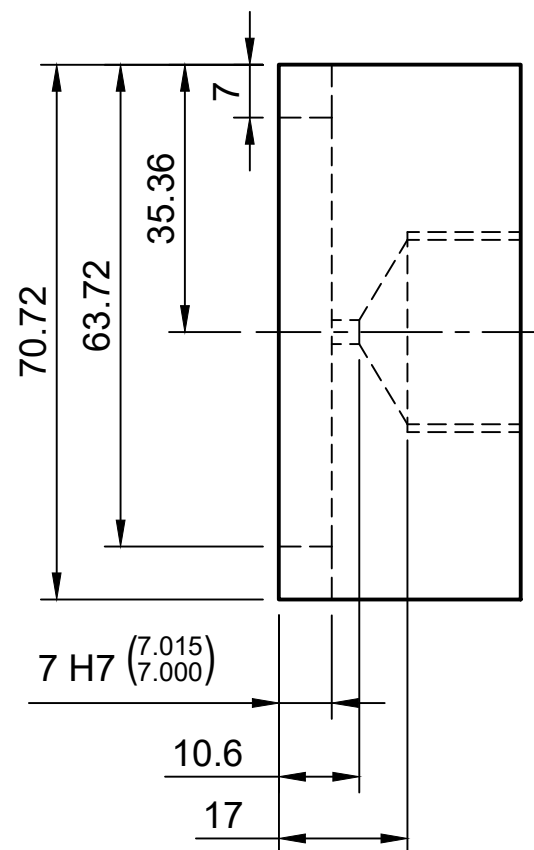
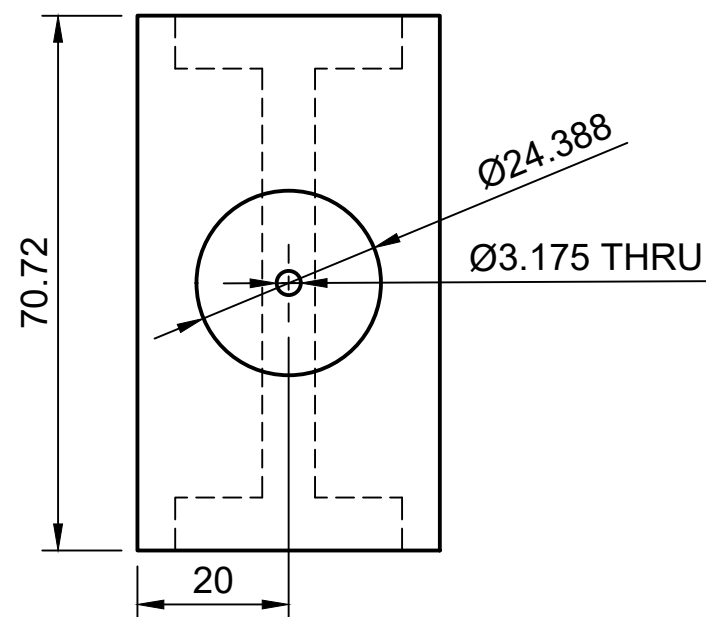
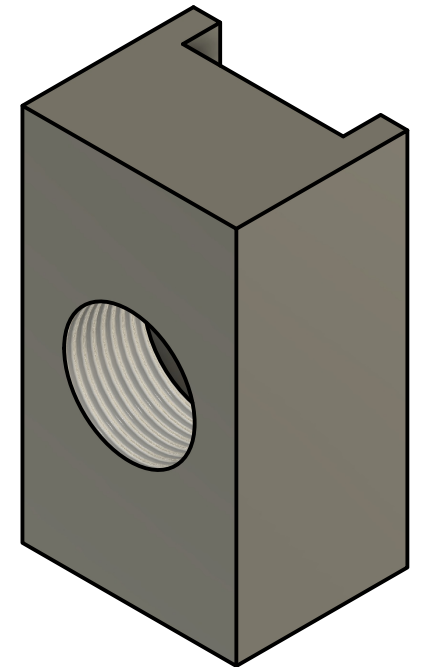


C-C (1:1)

7 H7 (7.015)
7.000



APPLY SURFACE FINISH TO
BOTH ADJACENT VERTICAL
FACES PER NOTE 1

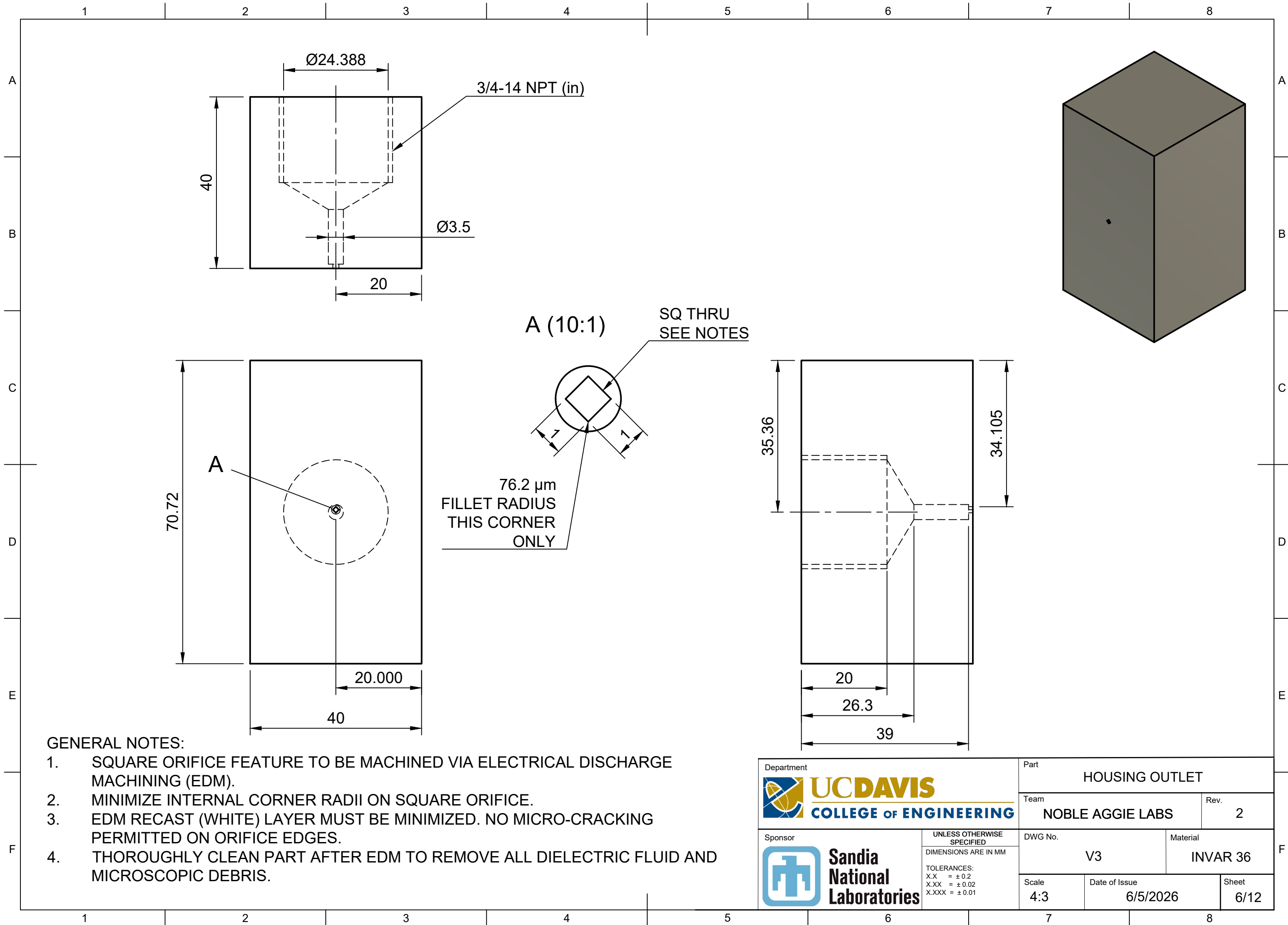


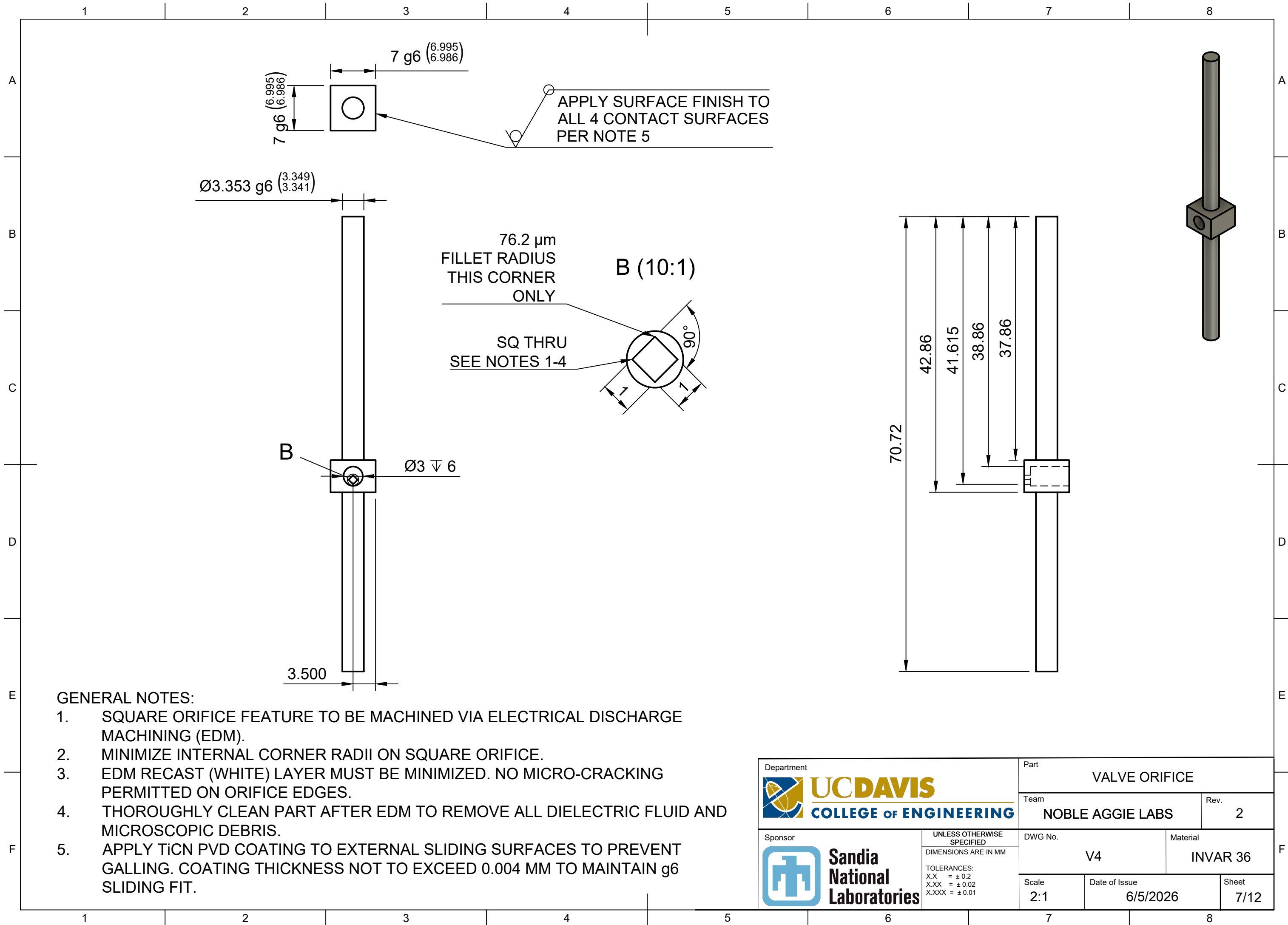
GENERAL NOTES

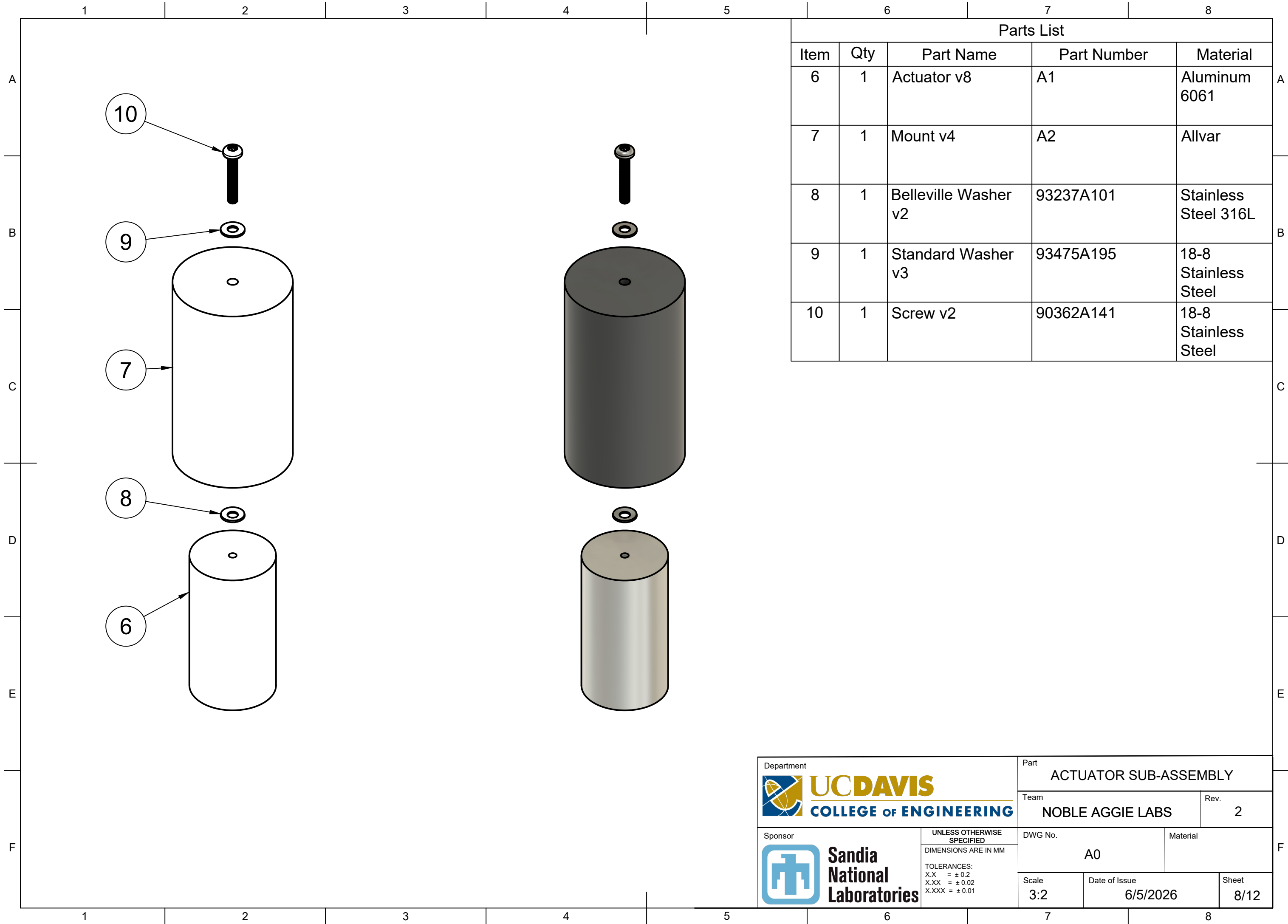
1. APPLY TiCN PVD COATING TO EXTERNAL SLIDING SURFACES TO PREVENT GALLING. COATING THICKNESS NOT TO EXCEED 0.004 MM TO MAINTAIN H7 SLIDING FIT.

Department UC DAVIS COLLEGE of ENGINEERING		Part HOUSING INLET	
Sponsor Sandia National Laboratories		Team NOBLE AGGIE LABS	Rev. 2
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM TOLERANCES: X.X = ±0.2 X.XX = ±0.02 X.XXX = ±0.01		DWG No. V2	Material INVAR 36
Scale 1:1		Date of Issue 6/5/2026	Sheet 5/12

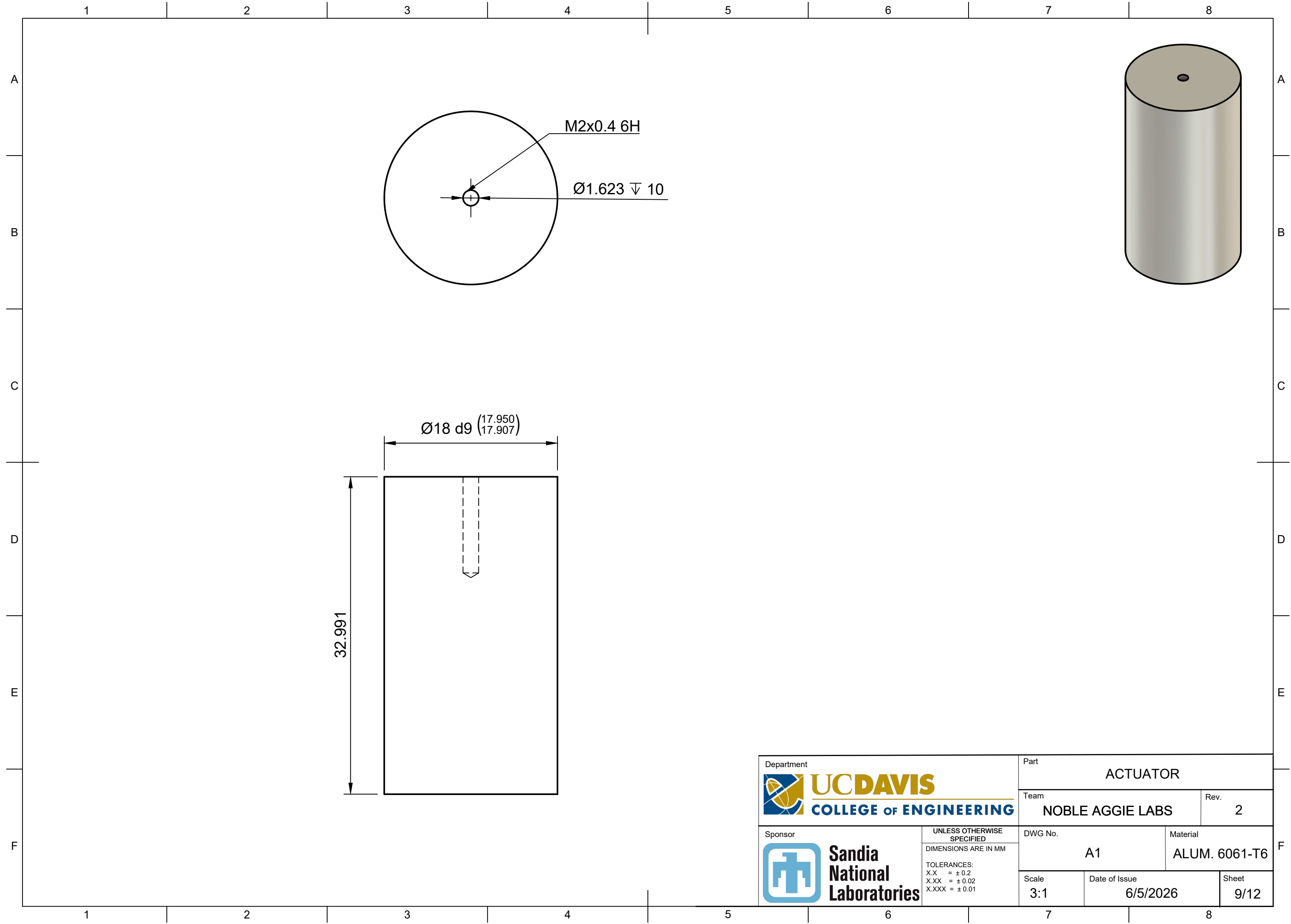
1 2 3 4 5 6 7 8

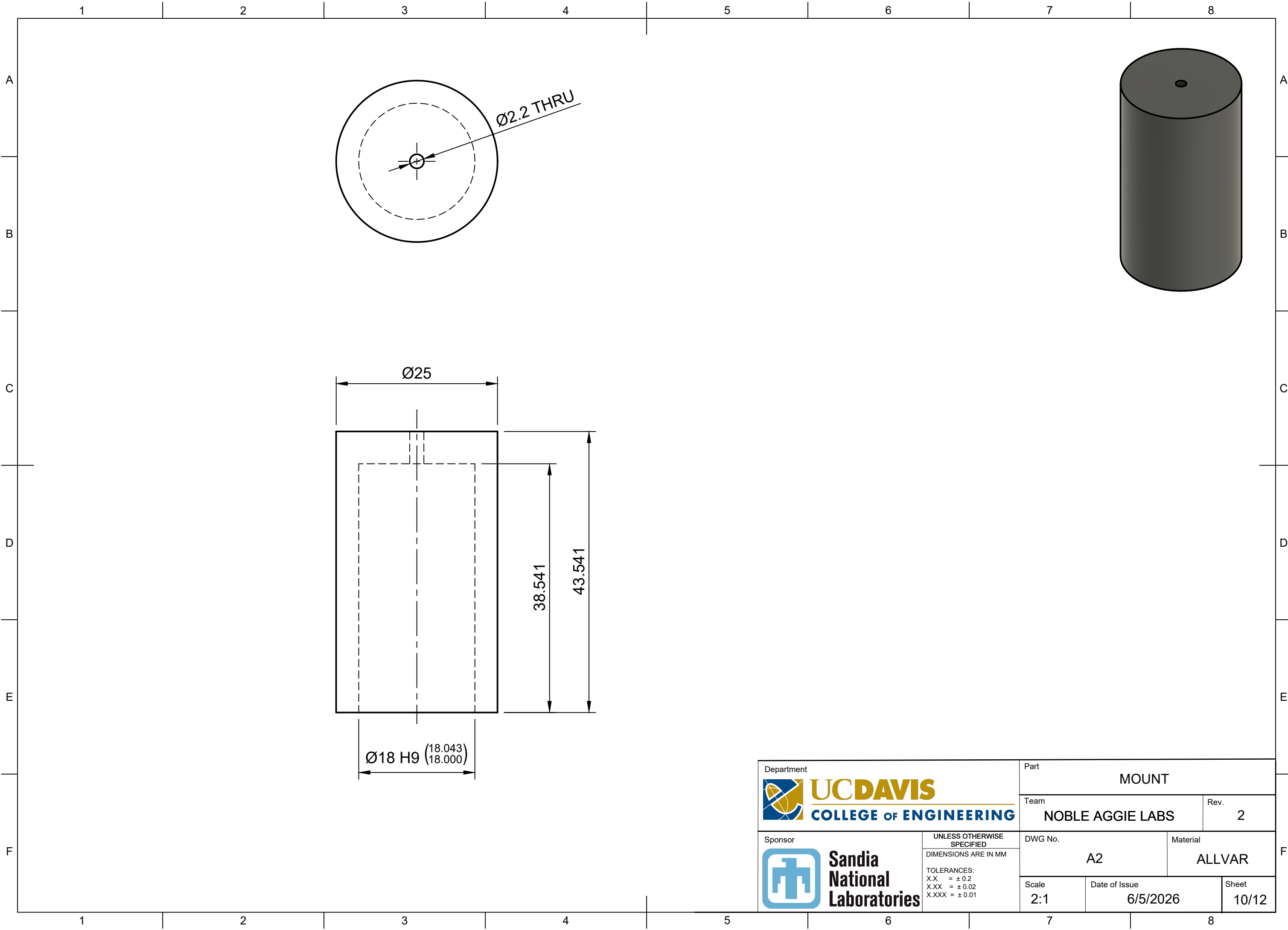




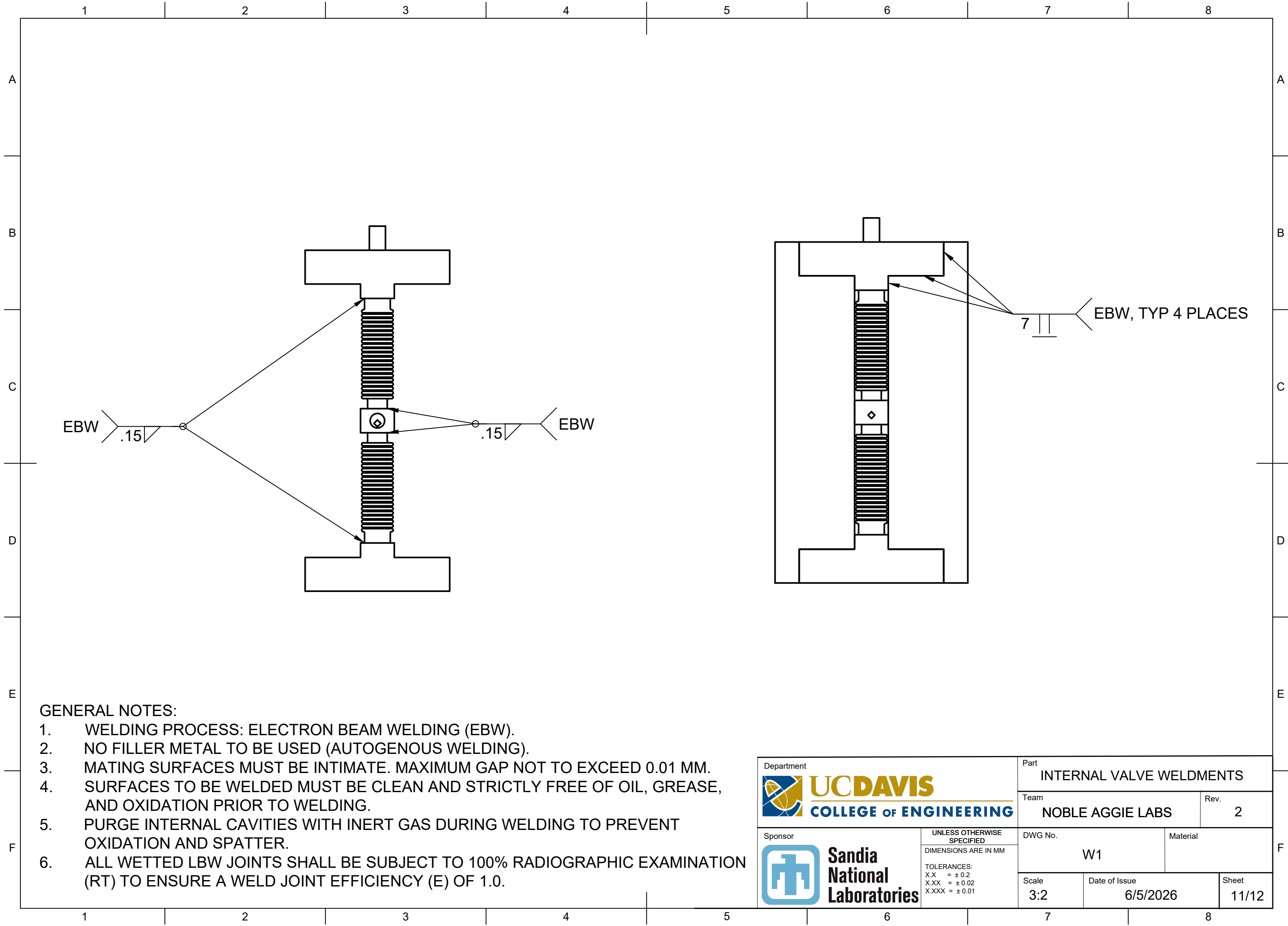


Department		Part		
		ACTUATOR SUB-ASSEMBLY		
Sponsor		Team		Rev.
		NOBLE AGGIE LABS		2
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.		Material
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		A0		
Scale		Date of Issue		Sheet
3:2		6/5/2026		8/12





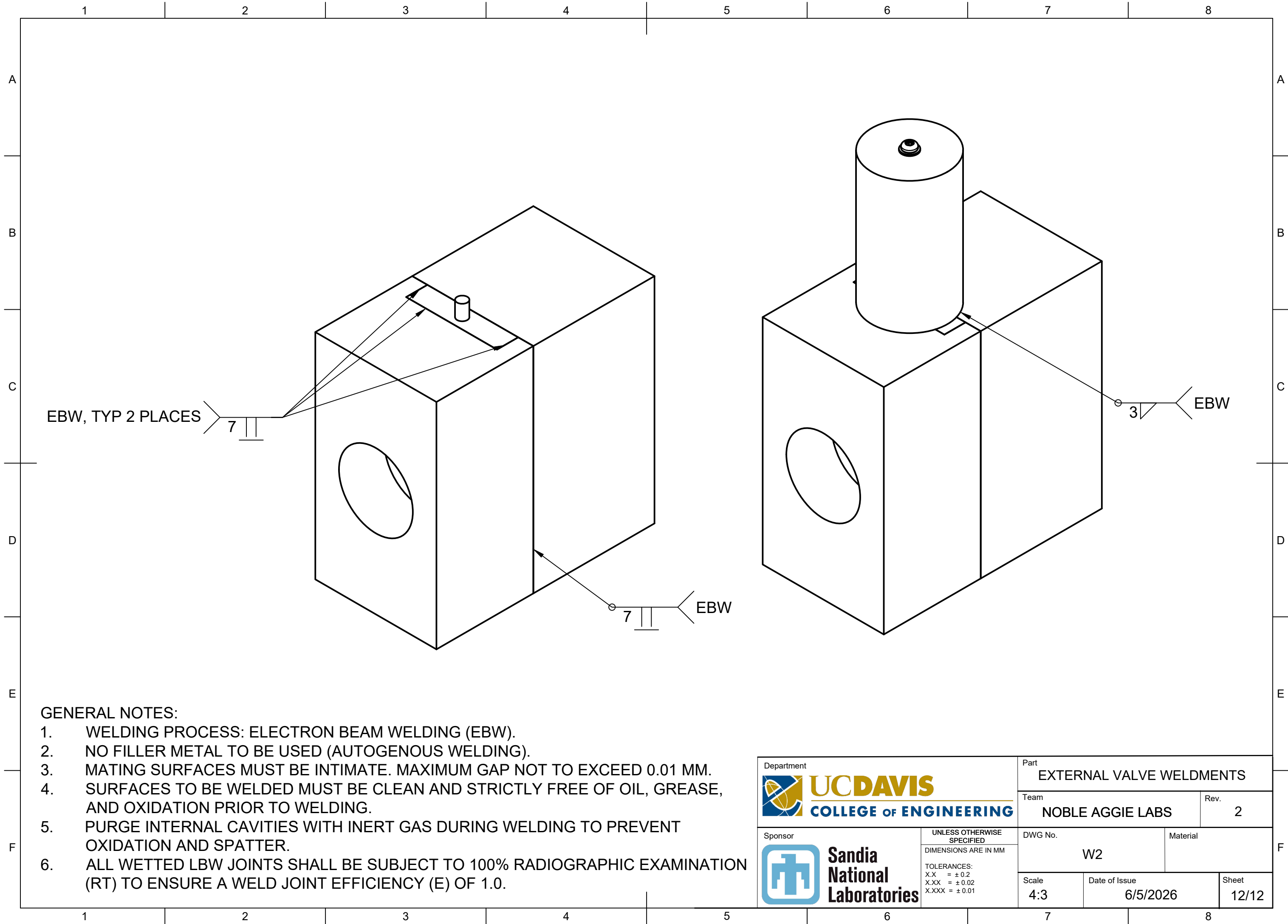
Department		Part		
		MOUNT		
Team		NOBLE AGGIE LABS		Rev. 2
Sponsor		DWG No.	Material	
		A2	ALLVAR	
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		Scale	Date of Issue	Sheet
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		2:1	6/5/2026	10/12



GENERAL NOTES:

1. WELDING PROCESS: ELECTRON BEAM WELDING (EBW).
2. NO FILLER METAL TO BE USED (AUTOGENOUS WELDING).
3. MATING SURFACES MUST BE INTIMATE. MAXIMUM GAP NOT TO EXCEED 0.01 MM.
4. SURFACES TO BE WELDED MUST BE CLEAN AND STRICTLY FREE OF OIL, GREASE, AND OXIDATION PRIOR TO WELDING.
5. PURGE INTERNAL CAVITIES WITH INERT GAS DURING WELDING TO PREVENT OXIDATION AND SPATTER.
6. ALL WETTED LBW JOINTS SHALL BE SUBJECT TO 100% RADIOGRAPHIC EXAMINATION (RT) TO ENSURE A WELD JOINT EFFICIENCY (E) OF 1.0.

Department		Part		
		INTERNAL VALVE WELDMENTS		
Sponsor		Team	Rev.	
		NOBLE AGGIE LABS	2	
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.	Material	
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		W1		
		Scale	Date of Issue	Sheet
		3:2	6/5/2026	11/12



GENERAL NOTES:

1. WELDING PROCESS: ELECTRON BEAM WELDING (EBW).
2. NO FILLER METAL TO BE USED (AUTOGENOUS WELDING).
3. MATING SURFACES MUST BE INTIMATE. MAXIMUM GAP NOT TO EXCEED 0.01 MM.
4. SURFACES TO BE WELDED MUST BE CLEAN AND STRICTLY FREE OF OIL, GREASE, AND OXIDATION PRIOR TO WELDING.
5. PURGE INTERNAL CAVITIES WITH INERT GAS DURING WELDING TO PREVENT OXIDATION AND SPATTER.
6. ALL WETTED LBW JOINTS SHALL BE SUBJECT TO 100% RADIOGRAPHIC EXAMINATION (RT) TO ENSURE A WELD JOINT EFFICIENCY (E) OF 1.0.

Department		Part	
		EXTERNAL VALVE WELDMENTS	
Sponsor		Team	Rev.
		NOBLE AGGIE LABS	2
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.	Material
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		W2	
		Scale	Date of Issue
		4:3	6/5/2026
			Sheet
			12/12